

MACHINE LEARNING

END-TO-END ROADMAP 

Complete Plan

Stage 1: Prerequisites (Foundation)

Mathematics

- **Linear Algebra:** Vectors, matrices, matrix multiplication, eigenvalues/eigenvectors
- **Probability & Statistics:** Mean, median, variance, distributions, Bayes' Theorem
- **Calculus (Basic):** Derivatives, gradients, chain rule (mainly for deep learning)

Programming

- **Python:** Variables, loops, functions, list comprehensions
- **Libraries:**
 - `NumPy` (arrays, vectorized operations)
 - `Pandas` (data manipulation)
 - `Matplotlib`, `Seaborn` (visualization)

Stage 2: Core Machine Learning Concepts

Supervised Learning

- **Regression:** Linear Regression, Ridge, Lasso
- **Classification:** Logistic Regression, KNN, Decision Trees, Random Forest, SVM

Model Evaluation

- Accuracy, Precision, Recall, F1 Score
- Confusion Matrix, ROC-AUC
- Cross-validation

Unsupervised Learning

- **Clustering:** K-Means, Hierarchical, DBSCAN
- **Dimensionality Reduction:** PCA, t-SNE

Tools

- **Scikit-learn:** ML algorithms and pipelines
- **XGBoost, LightGBM:** Advanced tree-based models

Stage 3: Model Deployment and Tools

- **Model Pipelines:** `scikit-learn.pipeline`, preprocessing
- **Deployment:**
 - Flask / FastAPI for creating APIs
 - Streamlit / Gradio for building ML apps
 - Docker for containerizing models
 - Cloud: AWS/GCP/Azure basics
- **Version Control:** Git, GitHub
- **Experiment Tracking:** MLflow, Weights & Biases (W&B)

Stage 4: Deep Learning (Neural Networks)

- **Neural Networks:** Perceptron, activation functions, backpropagation
- **Frameworks:** TensorFlow / PyTorch
- **CNNs:** Image classification

- **RNNs & LSTMs:** Time-series & NLP basics
- **Transformers:** BERT, GPT basics (for NLP)

Stage 5: Specialized Fields

NLP (Natural Language Processing)

- Tokenization, Embeddings (Word2Vec, GloVe)
- Sequence models: LSTM, GRU
- Transformers: BERT, GPT, Hugging Face

Reinforcement Learning

- Markov Decision Processes
- Q-learning, Deep Q Networks (DQN)

Time Series Analysis

- ARIMA, Prophet
- Feature engineering for time-based data

Stage 6: Projects & Portfolio

- **Ideas:**
 - House Price Prediction
 - Credit Card Fraud Detection
 - Customer Churn Analysis

- Face Recognition App
- Sentiment Analysis on Tweets
- Image Classifier using CNNs
- **Tips:**
 - Use GitHub to document projects
 - Write blog posts or create YouTube videos explaining projects
 - Deploy at least 2 projects (Streamlit / Flask)
 - Participate in Kaggle competitions

Bonus: Resources

Courses:

- Coursera: Andrew Ng's ML & Deep Learning Specialization
- fast.ai
- Kaggle Courses (free)

Books:

- *Hands-On ML with Scikit-Learn, Keras, and TensorFlow* — Aurélien Géron
- *Pattern Recognition and Machine Learning* — Christopher Bishop

